

- 2 A fixed mass of an ideal monatomic gas undergoes a cycle **ABCA** of changes, as shown in Fig. 2.1.

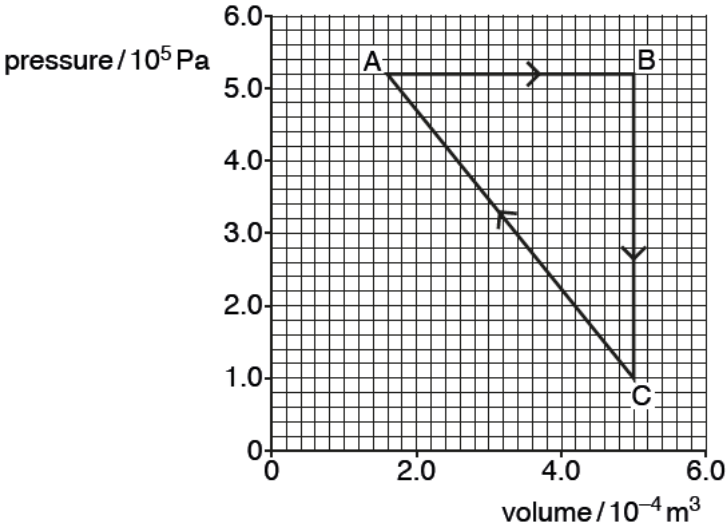


Fig. 2.1

- (a) During the change from **B** to **C**, the internal energy of the gas decreases by 315 J.

By considering molecular energy, state and explain qualitatively the change, if any, in the temperature of the gas.

..... There is negligible potential energy PE. As such, a decrease in internal
..... is equivalent to a decrease in kinetic energy. [B1]
..... Since KE is proportional to thermodynamic T, a decrease in kinetic
..... energy will lead to a decrease in the temperature of the gas. [B1]
..... [2]

- (b) During the change from **A** to **B**, the energy supplied to the gas by heating is 442 J.

Fig. 2.2 is a table of energy changes during one cycle. Complete Fig. 2.2.

section of cycle	heat supplied to gas / J	work done on gas / J	increase in internal energy of gas / J
A to B	442	-177	265
B to C	-315	0	-315
C to A	-55	105	50

Fig. 2.2

[3]

Since $\Delta U = 0$ for the entire cycle **ABCA**,
 ΔU (for process C to A) = $315 - 265.2 = 49.8$

Work done on gas (from C to A) = area under graph
 $= (5 - 1.6) \times 10^{-4} \times (5.2 + 1) / 2$
 $= 105.4 \text{ J}$

By First Law of Thermodynamics,

$$\Delta U = Q + W$$

$$\Delta Q = 49.8 - 105.4$$

$$\Delta Q = -55.6 \text{ J}$$

All W correct – 1 mark

Last column correct (add up to 0) – 1 mark

First Law correct – 1 mark

- (c) Calculate the root-mean-square speed of the gas atoms at point **B** if the root-mean-square speed at point **A** is 350 m s^{-1} .

Since $\frac{1}{2} \langle c^2 \rangle = \frac{3}{2} kT$

$\sqrt{\langle c^2 \rangle}$ proportional to $\sqrt{T}$ or $\sqrt{P}$ when P is constant

$$\frac{\sqrt{\langle c^2 \rangle}}{350} = \sqrt{\frac{5}{1.6}}$$

$$\sqrt{\langle c^2 \rangle} = 619 \text{ m s}^{-1}$$

root-mean-square speed = m s^{-1} [2]

2 (a) As seen in Fig 2.1, a monochromatic light of wavelength 500 nm is used to produce an